Minor Project - Equilibrium Analysis of a Cubic Nonlinear Spring

(23BME005)

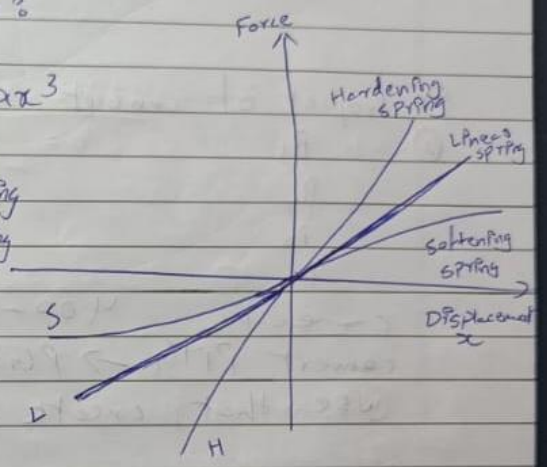
For Non-linear springs:

MASS-SPRING-DAMPER SYSTEM WITH A NONLINEAR SPRING

Springs with large deflections are not linear and their forces can be approximated by a cubic term:

$$F_s = -kx - ax^3$$

If $a > 0 \Rightarrow$ Hardening spring
If $a < 0 \Rightarrow$ softening spring



Hence, For a mass-spring system

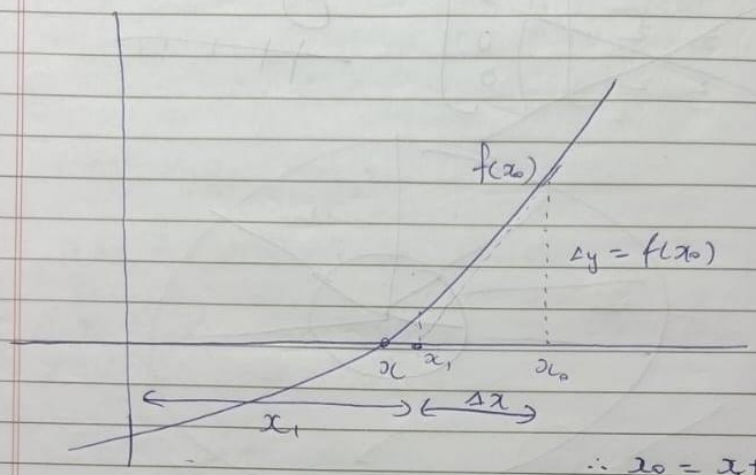
$$F = -kx - ax^3$$

$$\therefore M\ddot{x} + kx + ax^3 = 0$$

x_n1

$x_n = x_n1$

NEWTON - RAPHSON METHOD



$\therefore x_0 = x_1 + \Delta x$

$\therefore x_1 = x_0 - \Delta x$

$$\therefore \text{slope} = \frac{\Delta y}{\Delta x}$$

slope of $f(x)$ at x_0 $\rightarrow$ $\frac{d}{dx} f(x) = f'(x)$

$$\therefore f'(x)\Big|_{x=x_0} = f'(x_0)$$

$$f'(x_0) = \frac{f(x_0)}{\Delta x}$$

$$\therefore \Delta x = \frac{f(x_0)}{f'(x_0)}$$

$$\therefore x_1 = x_0 - \Delta x$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Hence, $\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$

Regula Falsi (False position)

$$f(a) \cdot f(b) < 0$$

instead of midpoint, approximate $f(x)$ by a straight line between $(a, f(a))$ and $(b, f(b))$

$$y - f(a) = \frac{f(b) - f(a)}{b - a} (x - a)$$

Find root,

$$y = 0$$

$$\therefore -f(a)(b-a) = (f(b) - f(a))(x-a)$$

$$\therefore -\frac{f(a)(b-a)}{f(b) - f(a)} = x - a$$

$$\therefore -\frac{f(a)(b-a)}{f(b) - f(a)} + a = x$$

$$\therefore x = a - \frac{f(a)(b-a)}{f(b) - f(a)}$$